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RATIONALITY OF PRELIMINARY MONEY STOCK ESTIMATES

Kenneth Kavajecz and Sean Collins*

Abstract—Earlier studies have presented mixed evidence on the rationality of the Federal Reserve's preliminary money stock estimates. We investigate the rationality of M1A, M1, M2, and M3 for both seasonally- and not seasonally-adjusted data. We find preliminary growth rates of these aggregates to be rational for not seasonally adjusted data but irrational when data are seasonally adjusted. Using Monte Carlo studies, we conclude that irrationality in seasonally adjusted data arises from the specific seasonal adjustment procedure used by the Federal Reserve. As a result, researchers conducting similar tests may want to focus exclusively on not seasonally adjusted data.

I. Introduction

IN this paper we revisit the rationality of preliminary estimates of the monetary aggregates published by the Federal Reserve. Rationality, in this context, means that the Federal Reserve's preliminary estimates of growth rates of the money stock are unbiased predictors of the growth rates of finally-revised data. Given the importance of the money stock in the conduct of monetary policy and in analyzing the state of the economy and given that policy makers must conduct monetary policy without the benefit of final estimates of the money stock it would seem important to ensure that preliminary estimates of the money stock are rational predictors.

Earlier studies have presented mixed evidence on this issue. Stone and Olson (1978), who use weekly data from 1971 through 1977, find that preliminary estimates of M1A are rational if data are not-seasonally adjusted (NSA), but irrational if seasonally adjusted (SA). Hein and Ott (1983) reach the same conclusion using monthly data for 1982. In contrast, Mankiw, Runkle, and Shapiro (1984) use quarterly data on M1A from 1954 to 1978 and find that preliminary money stock estimates are irrational. They attribute the irrationality to an errors-in-variables problem, which arises when the Federal Reserve omits relevant data

that would help forecast the "true" value of the money stock. More recently, Mork (1990) concludes that preliminary money stock estimates are irrational, but attributes the irrationality to inefficient forecasts. Both Mankiw et al. and Mork argue that decision makers should adjust preliminary estimates of the money stock to eliminate apparent biases.

In this paper we expand upon the literature by examining a broader array of monetary aggregates, as well as addressing apparent differences in results of past studies. In particular, we use monthly data from January 1965 to December 1978 to analyze M1A, and monthly data from March 1981 to March 1992 to analyze M1, M2 and M3. By considering both seasonally adjusted (SA) and not seasonally adjusted (NSA) estimates, we reach three conclusions.

First, we find that preliminary estimates of these four aggregates are essentially rational when data are NSA. The finding that preliminary NSA estimates of the money stock are rational is not too surprising. As we explain, the procedures used by the Federal Reserve to construct preliminary estimates of the money stock explicitly attempt to produce rational forecasts of the "true" NSA money stock. In contrast, we find that preliminary estimates of the monetary aggregates are irrational on an SA basis. This, too, is not particularly surprising, as the Federal Reserve's seasonal adjustment procedure purposely ignores much contemporaneously available information.

Second, we suggest that the strong rejection of rationality of M1A (NSA) found by Mankiw et al. and Mork may owe to a feature of their data; their preliminary and "true" data are not of the same frequency; that is, they are of different levels of temporal aggregation. Mixing of data frequencies likely biases rationality tests toward rejection of rationality, and when we correct their data for this difference in frequency preliminary M1A is a rational predictor of its final value.

Third, we demonstrate with Monte Carlo experiments that the seasonal adjustment procedure used by the Federal Reserve is prone to generate preliminary SA data that are irrational,

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even though the underlying NSA data are rational predictors. The Federal Reserve now forecasts preliminary seasonal factors for the monetary aggregates for the upcoming year *before* preliminary NSA data are observed. These preliminary seasonal factors are then held fixed for twelve months despite incoming data. We find that irrationality in preliminary SA data could, in large part, be eliminated by "concurrent seasonal adjustment": re-seasonally adjusting each time a new NSA observation becomes available. In any case, when assessing the rationality of preliminary data estimates, researchers should probably rely on not seasonally adjusted series.

II. Why Does the Money Stock Revise?

The Federal Reserve revises money stock estimates for at least four reasons. First, redefinitions occasionally lead to revised estimates. Second, the Federal Reserve occasionally receives corrections to data reported by depositories (commercial banks, thrifts, credit unions) or other financial institutions (money market mutual funds). Third, on a weekly basis the Federal Reserve collects data on the deposit liabilities of about 80% of all depositories. The deposit liabilities of the remaining 20% are collected less frequently, most often quarterly. In the interim, the liabilities of these less-frequently reporting institutions are estimated on a weekly basis by various methods. Moreover, some components used to construct the monetary aggregates are available only with long lags.¹ These components must be estimated until actual data become available, thus giving rise to potential revisions. Fourth, the Federal Reserve revises seasonal factors annually in order to take advantage of new NSA data that have come in over the last twelve months, and to take account of revisions to previously available NSA data, or to redefinitions.

All four sources of revisions may lead statistical tests to conclude that preliminary money stock figures are not rational predictors of their "true," or final, values. However, the third and fourth factors are arguably more important in terms of assessing the Fed's performance in constructing preliminary money stock estimates. Redefinitions that lead to rejection of rationality may be of less concern to the data collection agency because the

rationality tests are then comparing apples with oranges.² Corrections to data will also be of less concern to the data collection agency if they are relatively small and unsystematic, which tends to be the case for the monetary aggregates.

The third factor, estimating data for less-frequently reporting institutions and contemporaneously unobserved components of the monetary aggregates, is of prime concern. The Federal Reserve explicitly attempts to form reasonable predictions of these contemporaneously unavailable quantities. For example, until such data become available, the Fed estimates the deposit liabilities of smaller quarterly-reporting institutions by considering their statistical relationship with those of larger currently-reporting depositories. One would suspect that if these procedures were deficient, the Federal Reserve would take steps to correct them in order to ensure policy makers accurate monetary readings. The statistical evidence presented in this paper suggests the Federal Reserve's data collection procedures have indeed been successful, in the sense that its preliminary NSA estimates of the money stock are rational predictors of their final values.

The fourth factor, revisions to seasonal factors, may also be of some concern. By its very nature, the procedure currently used by the Federal Reserve to seasonally adjust the monetary aggregates ignores substantial contemporaneously available information. As noted, the Federal Reserve revises seasonal factors just once a year in February. The value for improving the seasonal adjustment of the monetary aggregates of any not-seasonally adjusted data that become available in the ensuing eleven months is ignored until the following February. Thus, the Fed's seasonal adjustment procedure could well lead one to find that preliminary SA estimates of the money stock are not rational predictors.

III. Empirical Framework

In testing the rationality of preliminary estimates of the monetary aggregates, we follow the theoretical framework laid out by Mankiw et al. (1984). An agency collects data and releases a

¹ For instance, IRA / Keogh accounts at thrifts and money market mutual funds.

² In our empirical work, we attempt to limit the effect of redefinitions by backing out revisions from the "true" data, insofar as possible. Mankiw et al. also note that redefinitions may be of limited concern if the effect of the redefinition is mainly on levels, because their tests of rationality (our's too) are based on growth rates.

preliminary estimate denoted m_t^p , of the “true” (final) value of some series m_t^f . Mankiw et al. consider two competing hypotheses of the construction of preliminary data.

Under Hypothesis 1, the preliminary data m_t^p are equal to the “true” value m_t^f plus a random error, which means the preliminary data have an errors-in-variables problem. If so, regressions of the form

$$m_t^p = \phi_0 + \phi_1 m_t^f + \mu_t \quad (1)$$

should find $\phi_0 = 0$ and $\phi_1 = 1$.

Under the null of the competing hypothesis, Hypothesis 2, the data collection agency uses raw data, and any other currently available information, to construct a preliminary, optimal, forecast of the “true” value m_t^f . In this case, the true value m_t^f is equal to the preliminary value m_t^p plus a random error. Thus, if Hypothesis 2 is correct, regressions of the form

$$m_t^f = \alpha + \beta m_t^p + \sum_{i=1}^k \gamma_i W_{it} + \epsilon_t \quad (2)$$

should find $\alpha = \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$ and $\beta = 1$ jointly, where the W_{it} are any contemporaneously available variables that may provide information about m_t^f .

Hypotheses 1 and 2 are mutually exclusive. Under Hypothesis 1, errors-in-variables in the preliminary data implies that they are irrational, or suboptimal, predictors of their true values. Econometrically, this is because the preliminary estimates, when conditioned on currently available information, are biased forecasts of the “true” values. Thus, there is some current information *not* being used by the data collection agency that would improve its preliminary estimates. If Hypothesis 2 is instead true, the data collection agency has formed an optimal forecast m_t^p of m_t^f by using all contemporaneously available information useful for predicting the true value m_t^f . Since the two hypotheses are mutually exclusive, we can use either (1) or (2) to test the rationality of the preliminary data. Following Mankiw et al., we choose to work with (2) rather than (1).

Equation (2) can be specified in levels or growth rates. Following Mankiw et al. and Mork, we estimate equation (2) in growth rate terms.³ Anticipating our empirical specification, we there-

fore modify the notation slightly:

$$g_t^f = \alpha + \beta g_t^p + \sum_{i=1}^k \gamma_i W_{it} + \epsilon_t \quad (2a)$$

where g_t^f and g_t^p are growth rates of “true” and preliminary money stock estimates.

Restrictions like $H_0: \alpha = \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$ and $\beta = 1$ in (2a) are standard fare in the rational expectations literature. In addition, rationality, as defined in that literature, typically requires the residuals $\hat{\epsilon}_t$ from regressions like (2a) to exhibit whiteness. To check this property, we report a battery of diagnostics, including a test for ARCH(1) errors, Ljung-Box Q statistics, and Breusch-Godfrey (BG) Lagrange multiplier tests of MA(3) and AR(3) errors. We also use Nyblom’s (1989) test for parameter constancy to check the stationarity of the residuals.⁴

IV. Data

The data are month averages of the four monetary aggregates M1A, M1, M2, and M3.⁵ These data are reported in the Federal Reserve Board Statistical Releases J.3 and H.6. Data for M1A

is usually assumed to have at least one unit root. Differencing eliminates the unit root that might otherwise lead to t - and F -statistics with nonstandard distributions. Second, levels of the preliminary money stock typically show breaks in February because of the Federal Reserve’s annual benchmark of the monetary aggregates. Differencing eliminates these breaks, which might otherwise bias the tests for rationality. Third, differencing reduces the likelihood that redefinitions will lead us to reject rationality.

⁴ We thank an anonymous referee for calling our attention to Nyblom’s test. Nyblom’s test has as its null the assumption of no parameter changes over the sample, with the alternative being that model parameters change according to a random walk (more precisely, a martingale). As noted by the referee, Nyblom’s test, when applied to the residuals in equation (2a), gives a test of stationarity of those residuals. The RATS code used to compute Nyblom’s statistic is available from the authors on request.

⁵ Month averages of the monetary aggregates are constructed by taking a weighted average of weekly values within a given month. Presently, weekly values for the deposits components of the monetary aggregates are constructed by taking simple averages of daily observations for weeks ending Monday (weekly values for money market mutual funds are one-day observations for Wednesdays). The weights assigned to weekly values in constructing month averages depend on the number of days that the week-ending Monday occupies in a given month. Thus, for most weeks, the weights are just 7 days divided by the number of days in that month. However, weeks-ending Monday that overlap months will differ. For example, the week ending Monday April 4, 1994 overlaps March and April 1994. Consequently, in constructing month-average M2 for April 1994, the weekly value for M2 for the week ending April 4, 1994 is assigned a weight of (4/30).

³ Justifications for estimating in growth rate terms rather than levels are three-fold. First, the level of the money stock

run from January 1965 to December 1978. Data for the other three aggregates cover March 1981 to March 1992. We use both seasonally adjusted and not-seasonally adjusted data. The first-published figures and previous month's revised figures are used to construct the first-published growth rate. We use the Federal Reserve's estimate of the monetary aggregates as of February 1994 as the "true" or final value of the money stock, and its growth from month to month as the "true" growth rate.⁶

The market variables we use are similar to those in Mankiw et al. We include in W_{it} the yield on the 3-month Treasury bill, the growth rate of the New York Stock Exchange index (NYSE 5000), and the slope of the yield curve—as measured by the spread between yields on the 30-year Treasury bond and the 3-month Treasury bill.⁷ Short-term interest rates are often thought to be an important determinant of money demand and may, therefore, help predict money growth. The other variables are included to capture market expectations about the strength of the economy, which, like the T-bill yield may indicate the direction of money growth. As Mankiw et al. note, a prime advantage of such market variables is that they are measured precisely, thus not complicating the test of rationality versus errors-in-variables. Finally, we include dummy variables for each month of the year in order to pick up any revisions that are seasonal in nature. In regression (2a), we constrain the seasonal dummies to sum to zero in order to avoid perfect collinearity with the constant α .

⁶ We have adjusted the preliminary estimates of M1A, M1, and M3 for definition changes that took place sometime after our preliminary data were published. The adjusted M1A series excludes from the first-published M1A series demand deposits held by foreign-related and official institutions, but includes non-bank travelers checks. The adjusted M1 series excludes from the first-published M1 series all thrift holdings of demand deposits and vault cash (1988) and includes all automatic transfer service (ATS) accounts at credit unions (1988). The adjusted M3 series is the first-published M3 series plus term eurodollars (1984). These adjustments left first-published M2 unchanged.

⁷ We include the *level* of the 3-month Treasury bill yield following Mankiw et al. As a referee pointed out, the level of interest rates may be non-stationary, thus possibly confusing our regression results, and also those of Mankiw et al. To determine whether the results to be presented below depend on the inclusion of the level of the (possibly) nonstationary Treasury bill yield, we also computed regressions with the *level* of the T-bill yield being replaced by its first *difference*. The two sets of regression results differ little, both qualitatively and quantitatively. As a result, we present only those results using the level of the 3-month Treasury bill yield.

V. Findings

By and large, we find that preliminary NSA estimates of the money stock are rational predictors of their "true" values, which accords with the results of Stone and Olsen (1978) and Ott and Hein (1983), but contrasts with those of Mankiw et al. (1984) and Mork (1990). For example, for the NSA monetary aggregate M1A, using quarterly data running from 1954:Q1 to 1978:Q4, Mankiw et al. find:⁸

$$g_t^f = .23 + .76 g_t^p, \\ (.111) \quad (.047)$$

$$DW = 2.7, \quad \bar{R}^2 = .72. \quad (3)$$

The F -statistic for testing $\alpha = 0$, $\beta = 1$ has a p -value of less than 0.001, indicating that the hypothesis $\alpha = 0$, $\beta = 1$ is rejected. On the other hand, using monthly data running from January 1965 to December 1978, we find for NSA M1A:

$$g_t^f = .02 + .99 g_t^p, \\ (.013) \quad (.008)$$

$$DW = 1.9, \quad \bar{R}^2 = .49. \quad (4)$$

The coefficients α and β are now much closer to 0 and 1, respectively, and the statistic $F(2,178) = 1.37$ for testing $\alpha = 0$, $\beta = 1$ has a p -value of 0.26, indicating that the null of rationality is now accepted.

Results for the monetary aggregates M1, M2 and M3 are presented in table 1. Broadly, the results indicate that the NSA aggregates (top third of table) are rational predictors of their "true" values for all three aggregates. Over the period January 1981 to March 1992, the estimates for α and β are rather close to 0 and 1. Formally, though, F -tests reject $H_0: \alpha = \gamma_1 = \gamma_2 = \dots = \gamma_k = 0$, $\beta = 1$ for M1 and M3, accepting it for M2 only. Inspection of the t -statistics indicates that the estimates of β for M1 and M3 are insignificantly different from unity, but the estimates of α lie slightly above zero.⁹ The bias in the constant may arise from our imperfect adjustments for definition changes. As evidence, we reestimated equation (2a) over the period January 1985 to March 1992 for M1, M2, and M3 because there were no important redefinitions after January 1985. These latter estimates for α

⁸ Standard errors in parentheses below coefficient.

⁹ The F -statistic for testing $\alpha = 0$, $\beta = 1$ (as opposed to $\alpha = \gamma_1 = \dots = \gamma_k = 0$, $\beta = 1$) for M1 is $F(21,78) = 1.26$ with a p -value of 0.29, thus accepting the hypothesis of rationality. Similarly, for M3 the p -value of the test $\alpha = 0$, $\beta = 1$ is 0.32.

TABLE 1.—REGRESSION RESULTS

Equation (2a) $g_t^f = \alpha + \beta g_t^p + \sum_{i=1}^k \gamma_i W_{it} + \epsilon_t$

Period	α	β	$\bar{R}^2$	Diagnostics (p -values except for DW)					Tests of Rationality (p -values)		
				DW	Q_{12}	$ARCH_1$	BG_3	Nyblom	$\alpha = 0, \beta = 1$	$\gamma_i = 0 \forall i$	$\alpha = 0, \beta = 1$ $\gamma_i = 0 \forall i$
NSA, 81:03–92:03											
M1	.06 (.034)	.99 (.011)	.99	1.9	.06	.00	.03	> .10	.29	.10	.00
M2	.07 (.037)	.97 (.017)	.98	1.6	.45	.30	.02	> .10	.04	.29	.08
M3	–.02 (.047)	.95 (.031)	.95	1.2	.00	.10	.00	> .10	.32	.07	.00
NSA, 85:01–92:03											
M1	–.01 (.030)	.99 (.008)	.99	1.5	.41	.02	.05	< .01	.51	.46	.07
M2	.01 (.064)	.94 (.023)	.99	1.7	.51	.67	.07	> .10	.02	.43	.12
M3	.15 (.080)	.95 (.031)	.95	1.4	.01	.00	.00	> .10	.06	.16	.08
SA, 81:03–92:03											
M1	.03 (.10)	.79 (.03)	.87	2.0	.03	.02	.02	> .10	.00	.00	.00
M2	.00 (.063)	.83 (.031)	.89	1.7	.00	.66	.02	> .10	.00	.02	.00
M3	–.16 (.072)	.75 (.043)	.84	1.4	.04	.30	.00	< .05	.00	.00	.00

and β , shown in the middle third of table 1, are again close to 0 and 1, respectively, but the *F*-statistics now accept the null hypothesis of rationality at the 5% level. Over either estimation period, the battery of tests for whiteness in the residuals, though not conclusive, generally favors the hypothesis that the residuals are white noise.

Results are quite different for seasonally adjusted data (bottom third of table 1). Estimates of α and β differ markedly from zero and one, and the *F*-statistics sharply reject the null of rationality. Rationality was still strongly rejected when we reestimated equation (2a) using seasonally adjusted data over the period January 1985 to March 1992 (results not reported). The diagnostic tests for whiteness in the residuals are ambiguous, with the tests for some aggregates confirming whiteness, but rejecting it for other aggregates.¹⁰

¹⁰ To test the sensitivity of the results to the possibility of a second unit root in money, we reproduced for both NSA and SA data the regressions in table 1 while imposing $\beta = 1$. If money has a second unit root and the restriction $\alpha = 0$, $\beta = 1$ is valid, the residual in the regression $(g_t^f - g_t^p) = \alpha + \epsilon_t$ should be stationary. A finding of a nonstationary residual might therefore raise suspicions about the appropriateness of the regressions in table 1. Following the suggestion of a referee, we computed such regressions. The diagnostic tests—in particular, the *Q* and Nyblom statistics—do not indicate that the residuals are non-stationary. Accordingly, we omit these regressions, but they are available on request.

We conclude that for all practical purposes preliminary estimates of the money stock are rational when data are not seasonally adjusted, but are irrational when data are seasonally adjusted.

VI. Comparing Results

Given that the results in the previous section differ markedly from those of Mankiw et al. (1984) and Mork (1990), an explanation is warranted. We conjecture that the strong rejection of rationality of NSA data found by Mankiw et al. and Mork owes to a feature of their data, namely, inconsistency between the dating of their preliminary and “true” money stock series.

Both articles take as their preliminary series figures published in Boschen and Grossman (1982). The Boschen-Grossman data are the stock of money at the end of the first month of each quarter.¹¹ However, for their “true” series, Mankiw et al. and Mork apparently use month-average data (rather than month-end) for the first

¹¹ Until 1960, the Boschen-Grossman data are the stock of M1A reported on the last Wednesday of the month. From 1960 to 1965, they are reported for an average of the days in the middle two weeks of the month. From 1965 to 1978 it is an average of the days of the middle week of the month.

month in each quarter. This pairing of month-end with month-average data biases tests of rationality toward rejection of rationality. The definition of rationality, $g_t^f = g_t^p + \epsilon_t$, requires the variance of the “true” series g_t^f to be greater than the variance of the preliminary series g_t^p , that is $\text{var}(g_t^f) > \text{var}(g_t^p)$. But in practice, money stock growth rates based on month-end data, g_t^w whether preliminary or “true,” will tend to have greater variance than growth-rates based on month-averages (g_t^m), that is $\text{var}(g_t^m) < \text{var}(g_t^w)$. Thus, if we pair growth rates based on month-average data with those based on month-end data and run the regression $g_t^m = \alpha + \beta g_t^w$, the hypothesis $\alpha = 0, \beta = 1$ will typically be rejected (example in appendix a). To demonstrate this, we computed the regression $g_t^m = \alpha + \beta g_t^w$ over the period 1965:01 to 1978:12, where g_t^m is the growth rate of month-average M1A, g_t^w is the growth rate of month-end M1A, and both g_t^m and g_t^w are based on the Federal Reserve’s current (“true”) estimate of M1A:

$$g_t^m = \begin{matrix} .15 \\ (.077) \end{matrix} + \begin{matrix} .69 \\ (.041) \end{matrix} g_t^w, \quad \bar{R}^2 = .64 \quad \text{and} \quad DW = 2.9. \quad (5)$$

The statistic for testing $\alpha = 0, \beta = 1$ is $F(2,166) = 27.20$, with a p -value of 0.00. The regression has no economic content, but the estimates are remarkably similar to those in (3), the result found by Mankiw et al. Since both they and Mork apparently use something closely akin to g_t^w as their preliminary series and use g_t^m as their “true” series it seems likely that their finding of irrationality in M1A (NSA) owes to inappropriate pairing of preliminary month-end data with “true” month-average data.

To check this, we replicated the regression of Mankiw et al. for the period 1961:Q1 to 1978:Q4.¹² The results are very similar to those of

Mankiw et al.:

$$g_t^f = \begin{matrix} .22 \\ (.141) \end{matrix} + \begin{matrix} .76 \\ (.058) \end{matrix} g_t^p, \quad \text{with } \bar{R}^2 = .70, DW = 2.7. \quad (6)$$

The statistic for testing $\alpha = 0, \beta = 1$ is $F(2,70) = 8.62$ with a p -value of 0.00, thus rejecting the null of rationality. Next, we corrected the analysis by pairing the preliminary growth rates g_t^p used by Mankiw et al. and Mork with the appropriate “true” growth rates g_t^f . For instance, if m_t^p (here m_t^p represents levels) in the Boschen-Grossman data set is for the last week in a month, we replace the month-average figure for m_t^f used by Mankiw et al. with a month-end value m_t^w for the last week in the same month. From these levels, we compute quarterly growth rates, g_t^{*f} . When we regress this corrected series of “true” growth rates, g_t^{*f} on the preliminary growth rates used by Mankiw et al. we find:

$$g_t^{*f} = \begin{matrix} .05 \\ (.092) \end{matrix} + \begin{matrix} .91 \\ (.039) \end{matrix} g_t^p, \quad \text{with } \bar{R}^2 = .88, DW = 2.1. \quad (7)$$

The F -statistic for testing $\alpha = 0, \beta = 1$ was $F(2,69) = 3.02$, with p -value = 0.06, accepting the null hypothesis of rationality at the 5% level.¹³ Therefore, when the preliminary M1A figures reported by Boschen and Grossman are paired with their appropriate final values, M1A is a rational predictor.

VII. Reconciling the Rationality of NSA Data with the Irrationality of SA Data

It is perhaps not too surprising that preliminary estimates of the money stock are rational for NSA data but irrational for SA data. One can easily prove that SA data will be irrational, even when based on rational NSA data, if the preliminary seasonal factors used to derive the SA data are irrational (proof in appendix b). There are a few reasons why preliminary seasonal factors might be irrational. For instance, in the case of money stock estimates, preliminary seasonal factors are based on lagged, rather than contemporaneous, data. Another possibility is that the procedure used to seasonally adjust the monetary aggregates—Census X11—may inadequately capture the seasonality in the monetary aggre-

¹² The preliminary money stock figures used by Mankiw et al. are reported in Boschen and Grossman. We did not have access to the “true” values used by Mankiw et al. We therefore matched, as nearly as possible, the Boschen-Grossman data with the Federal Reserve’s current estimate of *weekly* M1A. Because the Federal Reserve now maintains no official monetary series for dates before 1959, 1959 was as far back as we could carry the analysis. Also, we excluded figures before 1961 because we were unable to determine the precise week-ending dates of the Boschen-Grossman figures for 1959:Q1 to 1960:Q4.

¹³ The p -value rises to about 0.12 when we run the regression over the period 1965:Q1 to 1978:Q4.

gates.¹⁴ As a result, incoming data may lead seasonal factors, and in turn seasonally adjusted data, to revise predictably.

In order to investigate these possibilities we conducted two Monte Carlo experiments. For these experiments, we constructed ARIMA models whose NSA series are explicitly rational. We then ran both the preliminary and true NSA series through Census X-11, computed one-period growth rates for the resultant preliminary and true SA series, and estimated equation (2a). We conducted $j = 1$ to 500 trials, and by saving the simulated estimates $\hat{\alpha}_j$ and $\hat{\beta}_j$ (and the associated F -statistic for testing $\alpha = 0$ and $\beta = 1$, jointly) we can examine their distributions. If the estimated seasonal factors are “rational” forecasts of the true seasonal factors, we would expect the distributions of $\hat{\alpha}_j$ and $\hat{\beta}_j$ to be centered around zero and one, respectively.

For the experiments we chose two data generating processes (DGPs) which we hoped would offer verisimilitude. The first DGP is

$$\begin{aligned}(1 - B)(1 - B^{12})m_t^p &= (1 + .15B^6 - .8B^{12})u_t \\ u_t &\sim \text{iid } N(0, 1) \\ m_t^f &= m_t^p + \epsilon_t \\ \epsilon_t &\sim \text{iid } N(0, 1).\end{aligned}\quad (8)$$

This model, which has one unit root and one seasonal unit root, is taken from Pierce (1980), who notes that it describes the behavior of the monetary aggregate M1A reasonably well over the period 1969 to 1977. As constructed, the “true” series m_t^f is explicitly rational. The second DGP is

$$\begin{aligned}(1 - B)(1 - .25B)t_t &= 1 + u_t \\ u_t &\sim \text{iid } N(0, 1) \\ (1 - .5B^{12})s_t &= z_t \quad z_t \sim \text{iid } N(0, 1) \\ m_t^p &= t_t + s_t \\ m_t^f &= m_t^p + \epsilon_t \\ \epsilon_t &\sim \text{iid } N(0, 1).\end{aligned}\quad (9)$$

This second DGP is an unobserved components ARIMA model (UC-ARIMA) due to Engle

(1978). The preliminary data are composed of a trend t_t and a stationary stochastic seasonal s_t . As before, the “true” series is explicitly rational.

We assess the rationality of the seasonally adjusted data under two distinct cases. In the first case, we pass Census X-11 the *actual* preliminary data for both the estimation and the forecast horizon. We call this case “Case 1: Census X-11 with Actual Preliminary Data.” As noted above, the Federal Reserve forecasts preliminary seasonal factors up to twelve months in advance. Accordingly, in the second case we mimic the Federal Reserve’s seasonal adjustment algorithm by passing Census X-11 the actual preliminary data for the estimation period, *but twelve months of forecasts* from the DGPs (8) or (9). We call this case “Case 2: Census X-11-ARIMA.” These are polar cases in that Case 1 gives essentially “complete” information about underlying seasonality, whereas Case 2 is the least informative case likely to be seen in practice.¹⁵

Simulation results for the DGP (8) are in figure 1, which plots the distributions of $\hat{\alpha}_j$ and $\hat{\beta}_j$, as well as the distribution of the F -statistics for testing $\alpha = 0$ and $\beta = 1$, jointly. Solid lines are for Case 1, and dotted lines represent Case 2. The distributions of $\hat{\alpha}_j$ and $\hat{\beta}_j$ exhibit no skewness, and in both cases the means of $\hat{\alpha}_j$ are nearly zero. However, the distributions of $\hat{\beta}_j$ differ markedly between the two cases. For Case 1, the distribution of $\hat{\beta}_j$ is centered over unity, while for Case 2 the distribution is shifted to the left of unity. The F -statistics for testing $\alpha = 0$ and $\beta = 1$ (right-most box) show that the null hypothesis of rational SA data is rejected quite frequently with the X-11 ARIMA procedure (Case 2, dotted line), but falls to about the expected 5% level when we allow Census X-11 to use *actual* preliminary data (Case 1, solid line). Results for the DGP (9), shown in figure 2, are similar. The distribution of $\hat{\beta}_j$ is again shifted to the left of unity for the X-11 ARIMA procedure (dotted line). With the X-11 ARIMA procedure $\hat{\alpha}_j$ is now also biased, as its distribution lies significantly above zero. The irrationality in the SA data vanishes when Census X-11 has use of actual preliminary data. (Case 1, solid line). In sum, even though the “true” NSA

¹⁴ Census X-11 is an optimal seasonal adjustment procedure (optimal in the sense of minimizing the mean-square error of the seasonal adjustment) for certain kinds of ARIMA models. However, as pointed out by Pierce (1980), Census X-11 may not work as well for other kinds of models. The stochastic processes generating the monetary aggregates may or may not fall into the class of models for which Census X-11 works well.

¹⁵ Case 1 is obviously infeasible if the data collection agency insists on seasonally adjusting data well before final NSA data are known.

FIGURE 1.—MONTE CARLO DISTRIBUTIONS FOR DGP(8)

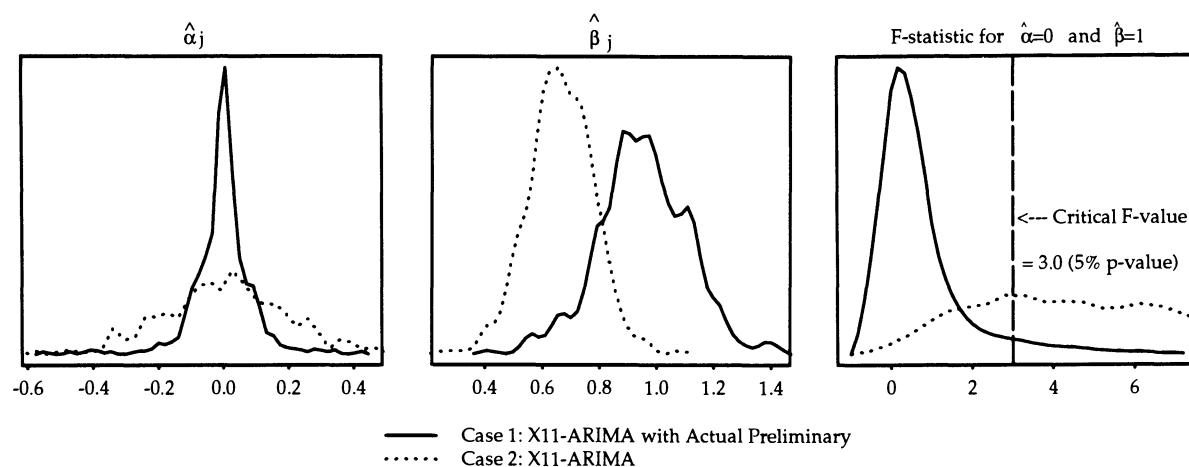
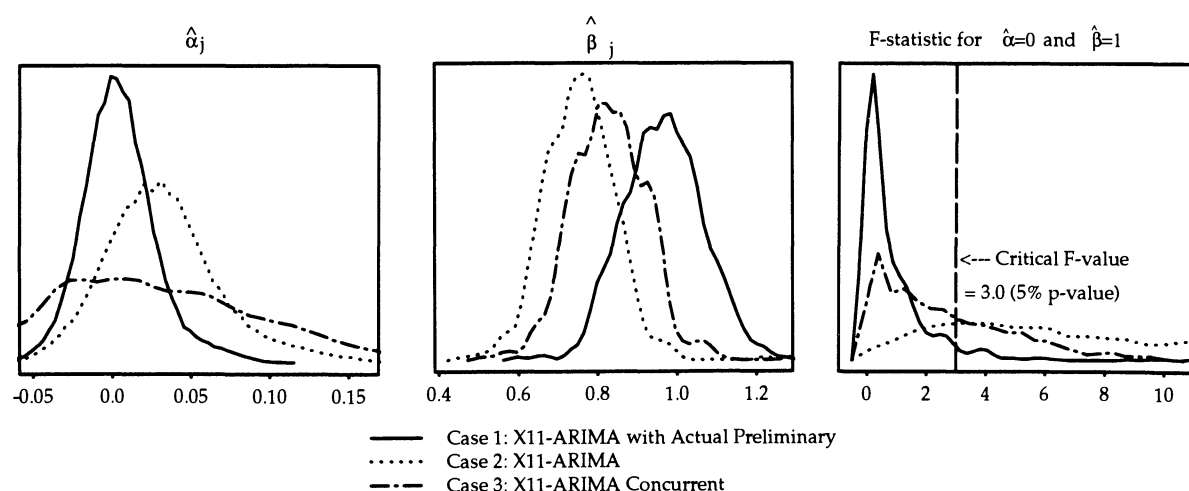


FIGURE 2.—MONTE CARLO DISTRIBUTIONS FOR DGP(9)



data are explicitly rational for both DGPs, Census X-11 ARIMA imparts irrationality to the SA data.

An intermediate alternative to our two polar cases is “concurrent” seasonal adjustment. Concurrent seasonal adjustment re-estimates seasonal factors each month in order to use the most recently available NSA data. It is an intermediate case in that seasonal factors are reestimated each month, rather than holding them fixed for twelve months.¹⁶ To investigate whether irrationality in

SA data can be eliminated by concurrent seasonal adjustment, we applied it to the DGP (9).¹⁷ The distributions for $\hat{\alpha}$ and $\hat{\beta}$ under the concurrent method are shown in figure 2 as “Case 3: X11-ARIMA Concurrent (dashed line).” Apparently, use of concurrent seasonal adjustment considerably improves the chances of concluding that seasonally adjusted data are rational. The distribution of the F -statistic for testing $\alpha = 0$ and $\beta = 1$ (right-most box in figure 2) is shifted substantially toward the solid line and away from the

¹⁶ Concurrent seasonal adjustment is always feasible for data collection agencies, but might be extremely costly. This may be why, as noted earlier, the Federal Reserve’s procedure is to reestimate seasonal factors just once a year.

¹⁷ We chose to apply the concurrent method to the DGP (9), because with (9) the distributions of both $\hat{\alpha}$ and $\hat{\beta}$ were biased allowing us to check for improvement in both of their distributions.

dotted line, indicating that the null hypothesis of rationality is now rejected much less frequently.¹⁸

VIII. Conclusion

By and large, the results here indicate that the Federal Reserve's preliminary estimates of the money stock are rational for not seasonally adjusted data. Mild departures from rationality that show up in the constant term may result from definition changes. In any case, remaining biases are small, and although statistically significant, are likely economically unimportant. Thus, there is little reason for policy makers to discount information in preliminary NSA money stock estimates.

On the other hand, preliminary estimates of seasonally adjusted money stock data are biased. However, we have shown that such biases may arise from projecting seasonal adjustment factors in advance of knowledge of preliminary NSA data. Therefore, while preliminary SA money stock data are irrational, the problem can, in part, be corrected by doing concurrent seasonal adjustment. It may be that in Monte Carlo experiments optimal seasonal adjustment procedures (Pierce, 1980) would completely alleviate the irrationality caused by seasonal adjustment. However, optimal seasonal adjustment procedures assume that the true underlying model is known, whereas in practice true models are unknown. In any case, the irrationality of the seasonally adjusted data cannot be corrected by collecting more or better information about the not seasonally adjusted monetary aggregates or their components.

Finally, researchers attempting to assess the rationality of other kinds of preliminary data may want to focus on not seasonally adjusted data. An independent example of the difficulty of interpreting rationality tests conducted with seasonally adjusted data is Neumark and Wascher (1991), who examine preliminary estimates of payroll employment. They conclude that preliminary estimates of payroll employment are irrational by analyzing two datasets. The datasets differ impor-

tantly because the first dataset applies the same seasonal factor to both the preliminary and first revised data, while the second dataset applies *different* seasonal factors to preliminary and final revised data. Although Neumark and Wascher rejected rationality for both datasets, the rejection is much stronger for the second dataset, which uses revised seasonal factors. This is indicative of our assertion that seasonal adjustment can seriously impact tests of rationality.

APPENDIX A

Example: Suppose the preliminary money stock is ${}_w x_t^p = \alpha + \epsilon_t$ with $\epsilon_t \sim \text{iid } N(0, \sigma_\epsilon^2)$, and the subscript w implies that ${}_w x_t^p$ is a weekly value. Let ${}_w x_t^f = {}_w x_t^p + u_t$, so that the preliminary weekly observations are rational predictors of their final weekly values. Let ${}_m \bar{x}^p = ({}_w x_t^p + {}_w x_{t-1}^p + {}_w x_{t-2}^p + {}_w x_{t-3}^p)/4$ be the month-average level of x computed by taking the simple average of the four weeks in a given month. Since ${}_m \bar{x}^f = {}_m \bar{x}^p + ({}_w u_1 + {}_w u_2 + {}_w u_3 + {}_w u_4)/4$, the preliminary month-average values are rational predictors of their final month-average values. But because of the dating of their data, the regression reported by Mankiw et al. is similar to running the regression:

$${}_m \bar{x}^f = \alpha + \beta {}_w x_t^p \quad (\text{A.1})$$

where ${}_w x_t^p$ is the last weekly observation in the month, and ${}_m \bar{x}^f$ is the month-average (Mankiw et al. do things in first differences, but the logic follows through). For rationality (i.e., $\alpha = 0$, $\beta = 1$) to hold statistically in (A.1) it must be that $\text{var}({}_m \bar{x}^f) > \text{var}({}_w x_t^p)$. But this condition will almost always be violated in practice because it reduces to the requirement that $\sigma_u^2 > 3\sigma_\epsilon^2$, or the variance of the revisions must be three times the variance of the money stock itself. In our data the variance of the money stock is about 100 times the variance of the revisions. Thus, even though the preliminary month-average values are rational, regressions like (A.1) will erroneously conclude they are not.

APPENDIX B

LEMMA: Assume the NSA series m_t^p is a rational predictor of a "true" NSA series m_t^f . Then the corresponding SA series ms_t^p is a rational predictor of the "true" SA series ms_t^f if and only if the preliminary seasonal factors s_t^p used to seasonally adjust m_t^p are a rational predictor of the "true" seasonal factors s_t^f used to seasonally adjust m_t^f .

Proof: The assumption that the preliminary NSA data are rational implies that the "true" NSA value m_t^f is equal to its preliminary estimate m_t^p plus a white noise disturbance.

$$m_t^f = m_t^p + \epsilon_t. \quad (\text{B.1})$$

Rationality of seasonal factors implies:

$$s_t^f = s_t^p + v_t \quad (\text{B.2})$$

where v_t is a white noise error. If we assume that not seasonally adjusted data are equal to a seasonally adjusted figure plus an additive seasonal, we have the additional two

¹⁸ The Federal Reserve published concurrent seasonally adjusted M1 from July 1982 to December 1989. Parameter estimates shifted in a similar way when we re-ran equation (2a) using the Federal Reserve's concurrent seasonally adjusted M1 series as preliminary data.

formulas:

$$m_t^p = ms_t^p + s_t^p, \text{ and } m_t^f = ms_t^f + s_t^f. \quad (\text{B.3})$$

Now consider the regression of ms_t^f on ms_t^p :

$$ms_t^f = \alpha + \beta ms_t^p + \omega_t. \quad (\text{B.4})$$

Substituting (B.1), (B.2) and (B.3) into (B.4) gives

$$(m_t^f - s_t^f) = \alpha + \beta(m_t^f - s_t^f) + (\omega_t - \epsilon_t + v_t). \quad (\text{B.5})$$

Since (B.5) is an identity, apart from the error, hypothesis tests based on regression estimates will accept $\alpha = 0$ and $\beta = 1$.

Conversely, if $\alpha = 0$ and $\beta = 1$ in (B.4), then substituting (B.1) and (B.3) into (B.4) and reducing gives:

$$s_t^f = s_t^p + (\omega_t - \epsilon_t). \quad (\text{B.6})$$

Therefore, the regression $s_t^f = \delta_0 + \delta_1 s_t^p$ will indicate that $\delta_0 = 0$ and $\delta_1 = 1$. QED.

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